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BUILD WITH भारत 4.0

NATIONAL LEVEL HACKATHON

TEAM NAME: Parikalp

PROBLEM STATEMENT: From satellite signals to actionable environmental hotspots

TEAM MEMBERS: Pratham Mittal,Priya Tomar,Priyanshi Dhama,Raghav Charaya, Arya Pandey

COLLEGE: ABES Institute Of Technology

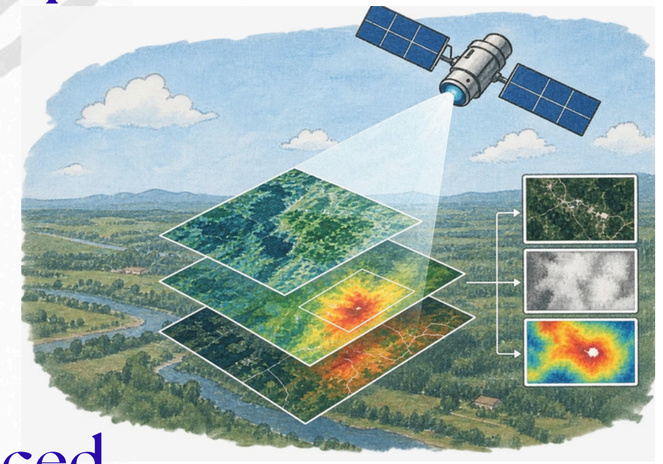
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PROBLEM STATEMENT

Fragmented Monitoring: Environmental signals analyzed separately

Unreliable Alerts: Clouds and seasons cause false alarms.



Limited Explainability: Alerts lack supporting evidence.

No Emerging Detection: Gradual problems remain unnoticed.

No Continuous Validation: Hotspots lack human monitoring.

Key Challenge | “What changed?” → “Is it reliable, emerging, and why?”

SOLUTION

ENVIRONMENTAL HOTSPOT INTELLIGENCE

An intelligence layer that detects, verifies and explains emerging environmental hotspots.



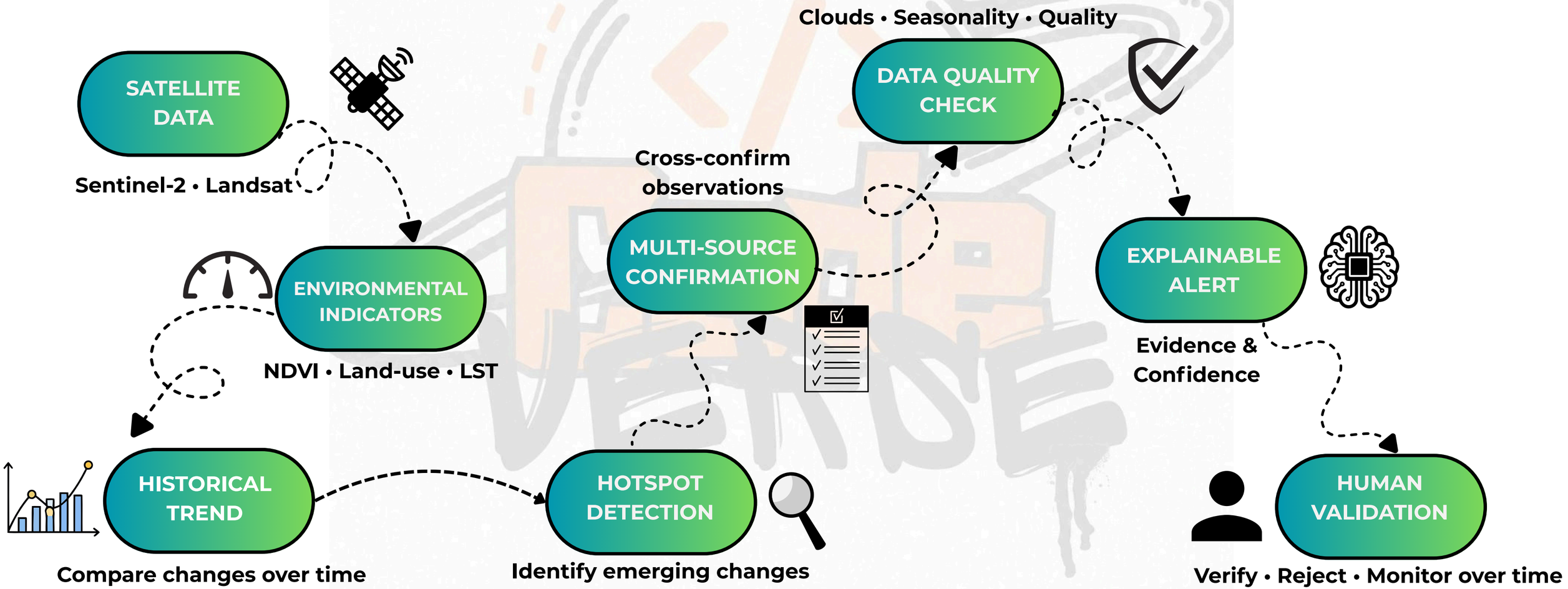
Core Approach

Detect → Confirm → Filter → Explain → Validate & Monitor

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FLOW OF SOLUTION



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TECH STACK

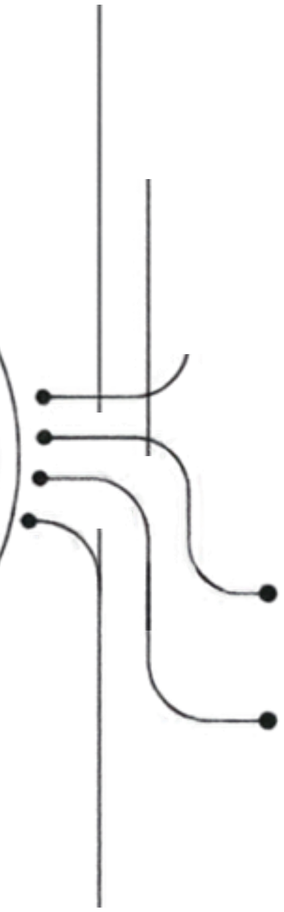
04 GOOGLE EARTH ENGINE
Satellite data & preprocessing

03 PYTHON
Hotspot/trend intelligence

02 FAST API
Backend & APIs

01 REACT
Interactive dashboard & maps

SYSTEM ARCHI TECTURE



SATELLITE DATA

GOOGLE EARTH ENGINE
Preprocessing + Environmental Indicators

PYTHON
Detect → Verify → Score → Explain

FASTAPI + POSTGIS
Backend + Geospatial Storage

REACT DASHBOARD
Map + Evidence + Trends

HUMAN VALIDATION
Confirm / Reject / Verify

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UNIQUE SELLING PROPOSITION (USP)



01
Detects gradual environmental deterioration

02
Verifies alerts using multiple observations

03
Shows indicators behind each alert

04
Enables human verification and monitoring

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FEASIBILITY & COMPETITORS

COMPETITORS

EXISTING PLATFORMS	THEIR FOCUS	OUR FOCUS
Google Earth Engine earthengine.google.com	Large-scale satellite data analysis	Integrated environmental hotspot detection
Global Forest Watch UNEP - UN Environment Programme	Forest monitoring and loss alerts	Combines several environmental indicators, not only forest loss
NASA Worldview UNEP - UN Environment Programme	Satellite imagery visualization	Flags emerging hotspots and presents evidence for review

FEASIBILITY

	FEASIBILITY	WHY?
SATELLITE DATA RETRIEVAL	🟢 High	GEE provides datasets and processing
NDVI / VEGETATION ANALYSIS	🟢 High	Established vegetation indicator
LAND-USE / BUILT-UP CHANGE	🟢 High	Existing classification methods
LST / TEMPERATURE ANALYSIS	🟢 High	Established satellite-derived indicator
DATA-QUALITY FILTERING	🟢 High	Cloud and seasonal checks

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RESEARCH & REFERENCE

A REVIEW ON CHANGE DETECTION METHOD AND ACCURACY ASSESSMENT (2021)

Satellite imagery + GIS are widely used to monitor land-use/land-cover change. Different change-detection methods are suitable

01

CHANGE DETECTION USING LANDSAT TIME SERIES: A REVIEW

Time-series satellite data can detect both abrupt and gradual changes. Seasonal differences can create noise/false changes.

02

COMPREHENSIVE REVIEW OF LULC AND LST (2024)

NDVI, NDBI and LST are commonly used together to study relationships between land-cover change and surface temperature.

03

RECENT APPLICATIONS OF LANDSAT 8 & SENTINEL-2

Landsat and Sentinel-2 can support LULC classification and change detection; combining sources can improve analysis.

04